USN | | | | | | | | | | | [22MTCSE301]

THIRD SEMESTER M.Tech EXAMINATION, DECEMBER - 2024

INFERENTIAL STATISTICS USING R

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
ii. Missing data may be suitably assumed.

1. a. Draw the two ogives for the following data, locate the median graphically and verify the median value. — CO1 — L3 — **10**

Weekly wages	0-10	10-20	20-30	30-40	40-50	50-60
No. of workers	3	9	15	30	18	5

b. For the following frequency distribution, find the mean, median and mode. — CO1 — L3 — **10**

Marks	0-4	4-8	8-12	12-16	16-20
No. of students	10	12	18	7	5

2. a. Define — CO2 — L2 — **12**

i. sampling and explain the methods of sampling with suitable examples.
ii. Null hypothesis and
iii. Alternative hypothesis.

b. If 2% of electric bulbs manufactured by a certain company are defective. Find the probability that in a sample of 200 bulbs — CO2 — L3 — **08**

i. less than 2 bulbs
ii. more than 3 bulbs are defective. Also mention which distribution will be suitable to solve the problem and why?

3. a. An electrical firm manufactures light bulbs that have a length of life that is approximately normally distributed with a mean of 800 hours and a standard deviation of 40 hours. Test the hypothesis H_0: $\mu = 800$ hours against H_1: $\mu \neq 800$, if a random sample of 30 bulbs has an average life of 788 hours. Use 5% and 1% level of significance. Find the 95% confidence interval for the population mean of all bulbs produced by this firm. ($Z_{0.025} = 1.96$, $Z_{0.005} = 2.58$) — CO3 — L3 — **10**

b. Two random samples were drawn from two normal populations and their values are — CO3 — L3 — **10**

A: 16, 17, 25, 26, 32, 34, 38, 40, 42

B: 14, 16, 24, 28, 32, 35, 37, 42, 43, 45, 47

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		Test whether the variance of A is lesser than B at 5% level of significance. ($F_{0.05}(10,8) = 3.35$)			
		OR			
4.	**a.**	Following table gives the number of breakdowns in a factory on various days of a week.	CO4	L3	**12**

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
No. of breakdowns	14	22	16	18	12	19	11

Using Chi-square test, check whether breakdowns are uniformly distributed or not. ($\chi^2_{\alpha} = 12.592$)

	b.	Explain the differences between Parametric and Non-Parametric tests with examples.	CO4	L3	**08**
5.	**a.**	Following is an arrangement of 25 men "M" and 15 women "W", lined up to purchase tickets for a premier picture show: MWWMMMWMMWMWMWWWMMMWMMWWWMMMMMMWWWMMMMMM. Test for randomness at 5% level of significance. ($Z_{0.025} = 1.96$)	CO4	L3	**10**
	b.	Write the procedure of sign test and Wilcoxon signed rank test. Also differentiate between the two.	CO4	L3	**10**
		OR			
6.	**a.**	For the following data, test the hypothesis that the median measure in the population X is less than the median measure in the population Y, using Mann-Whitney U-test at 5% level of significance. Given that the critical value of U for $n_1 = 4$, $n_2 = 7$ at 5% level of significance is 4. X: 60 45 23 32 Y: 10 25 20 54 32 65 08	CO5	L3	**10**
	b.	Reaction time for three drugs is given below. Using Friedman test, test the claim at the 5% significance level that all three drugs have the same probability distribution. ($\chi^2_{0.05,2} = 5.99$)	CO5	L3	**10**

Drug A	1.21	1.63	1.42	2.43	1.16	1.94
Drug B	1.48	1.85	2.06	1.98	1.27	2.44
Drug C	1.56	2.01	1.70	2.64	1.48	2.81

7.	**a.**	Calculate the SS_{total} for the following RBD data:	CO6	L3	**04**

	Scale		
Inspector	I	II	III
1	16	10	11
2	15	9	14
3	13	11	13

	b.	The three samples given below have been obtained from normal populations with equal variances. Test the hypothesis that the sample	CO6	L3	**16**

means are equal, by analysis of variance for the given data. ($F_{0.05}$ = 3.88)

A	B	C
8	7	12
10	5	9
7	10	13
14	9	12
11	9	14

OR

8. a. Following data represents per hectare yield for three varieties of wheat each grown in four plots. Using one-way ANOVA table. Test whether there is a significant difference between the average yields in the three varieties of wheat. ($F_{0.05}$ (2,9) = 4.26) — CO6 — L3 — **10**

Variety of Wheat		
A	B	C
6	5	5
7	5	4
3	3	3
8	7	4

b. Analyze the variance in the following Latin square of yields (in kgs) of paddy where A, B, C, D denote the different methods of cultivation. ($F_{0.05}$ (3,6) = 4.76) — CO6 — L3 — **10**

D 122	A 121	C 123	B 122
B 124	C 123	A 122	D 125
A 120	B 119	D 120	C 121
C 122	D 123	B 121	A 122

Examine whether the different methods of cultivation have given significantly different yields.